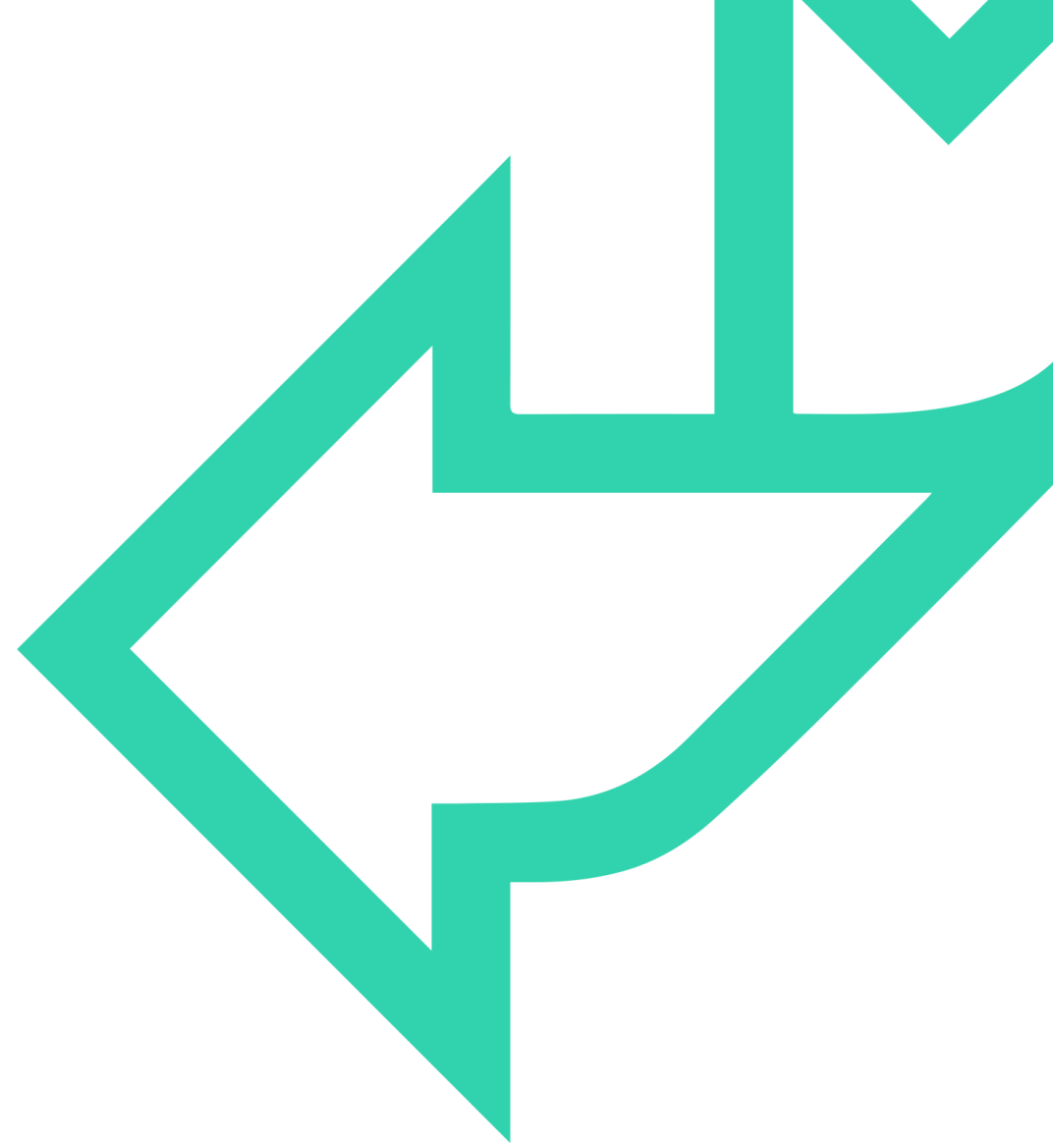




Control Flow - Selection





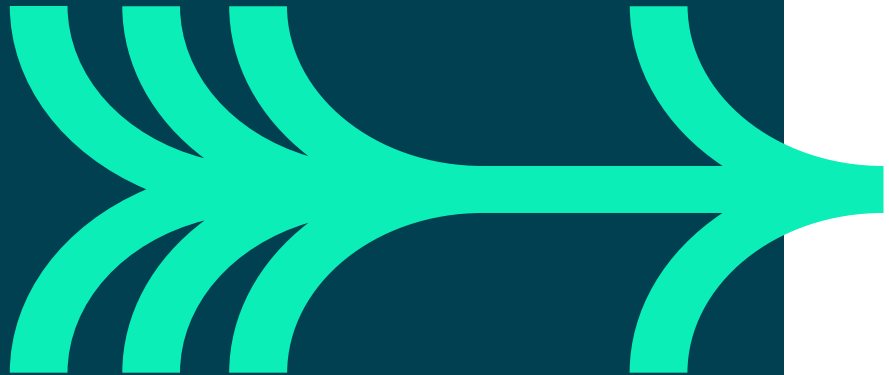
LESSON OBJECTIVES

- In this chapter, you'll learn about:
 - Python Control Flow statements
 - if
 - else
 - elif
 - Logical **OR** and **AND** operators





WHAT IS CONTROL FLOW?



Control Flow controls the order in which we do things:

- **Sequence**
 - Running code step by step, in order
- **Selection**
 - Deciding which lines of code should run
- **Iteration**
 - Doing the same thing many times, i.e. in a loop



CONTROL FLOW - SELECTION

Making decisions

if: do this

else: do something else





COMPARING VALUES

Relational operators for making a selection

>	Greater than
>=	Greater than or equal to
<	Less than
<=	Less than or equal to
==	Equal to comparison
!=	Not equal to comparison

Note!

== not the same as =

SELECTION EXAMPLE – IF STATEMENT



x = 5

Condition
Is x equal to 5?

if x == 5:
↔ print("X equals 5")

Yes, it is
... so do this



Use the Tab key to include
lines in an IF statement



SELECTION EXAMPLE – IF ... ELSE

X = 5

Is x equal to 5?

```
if x == 5:
```

```
    print("X equals 5")
```

```
else:
```

```
    print("X does not equal 5")
```

Yes, it is...
so do this

No it isn't
... so do this
instead



CONTROL FLOW USING 'ELSE IF' - ELIF



```
salary = 2500
```

```
if salary > 100000:  
    print('Band A')  
elif salary > 55000:  
    print('Band B')  
elif salary > 32000:  
    print('Band C')  
elif salary > 25000:  
    print('Band D')  
else:  
    print('Band E')
```

Creates a Chain of Tests

When a test passes, the other tests will not execute and the statement ends.

The last **else** is optional



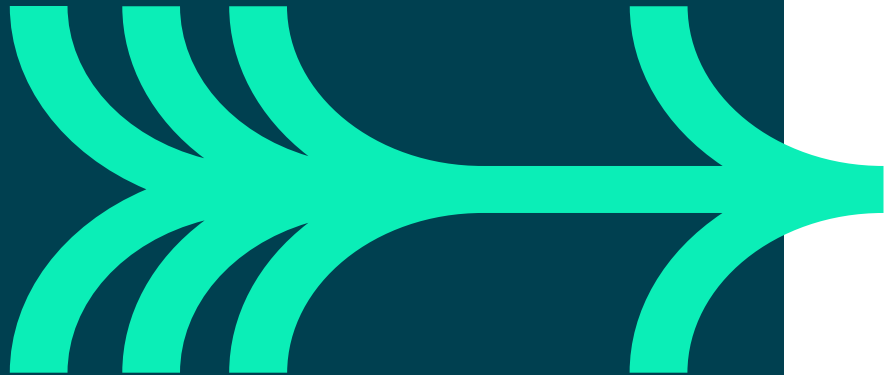
LOGICAL OR AND AND

What Would You Conclude From the Following Two **IF** Statements?

```
course='Python'  
age = 19
```

```
if course == 'Python' and age > 18:  
    print('Welcome!')
```

```
if course == 'Python' or age > 20:  
    print('Please start your', course, 'course!')
```





PYTHON FUNDAMENTALS



In this chapter, you learnt about:

- Python control flow statements
 - if
 - else
 - elif
- Logical **OR** and **AND** operators



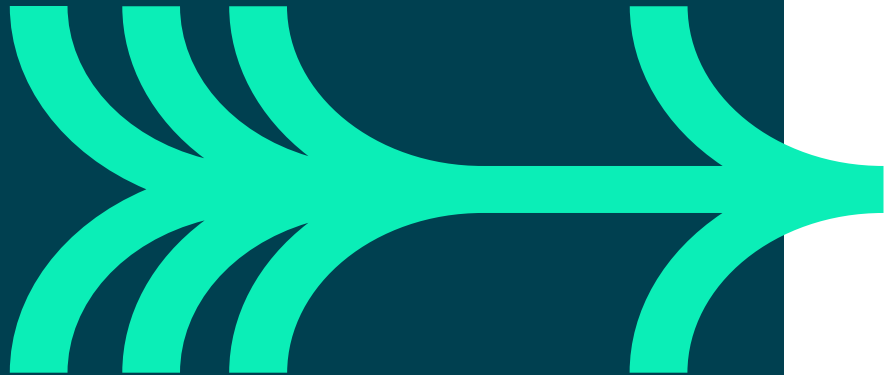
Guided Task

→ Please open the instructions in Cloud Academy



FURTHER READING

- <https://www.python.org/>
- <https://www.python.org/dev/peps/pep-0008/#a-foolish-consistency-is-the-hobgoblin-of-little-minds>





Thank you!

Hope you enjoyed this learning journey